

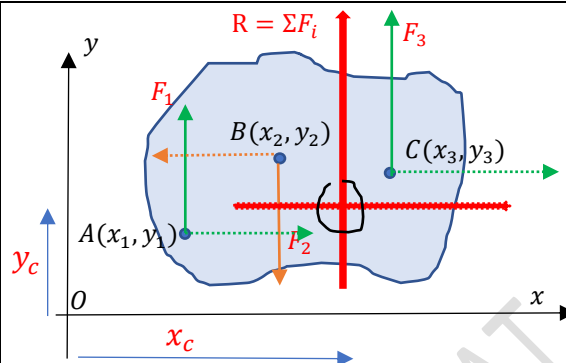
Centre of Parallel forces and centre of gravity

Consider a system of parallel forces $F_1, F_2, F_3 \dots$ acting at points $A(x_1, y_1), B(x_2, y_2), C(x_3, y_3) \dots$. The resultant of this system of parallel forces is $R = F_1 + F_2 + F_3 + \dots = \Sigma F_i$

The line of action of this resultant force can be obtained by applying the Varignon's theorem with respect to the origin O.

$$R \times x_c = F_1 \times x_1 + F_2 \times x_2 + F_3 \times x_3 + \dots$$

$$x_c = \frac{\Sigma F_i x_i}{\Sigma F_i}$$



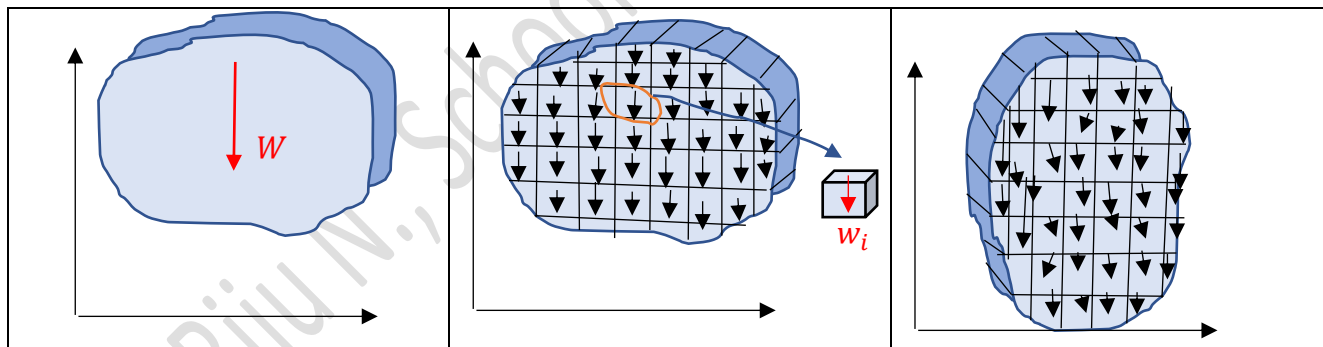
Now, let the force system be rotated through an angle 90° clockwise. The resultant also get rotated through 90° . The line of action of this resultant force can be obtained by applying the Varignon's theorem again.

$$-R \times y_c = -F_1 \times y_1 + F_2 \times y_2 - F_3 \times y_3 + \dots$$

$$y_c = \frac{\Sigma F_i y_i}{\Sigma F_i}$$

Irrespective of the orientation of the parallel forces in the plane, their resultant will be acting through a unique point whose co-ordinates are (x_c, y_c) . This point is known as Centre of parallel forces.

Consider a block of weight W having small uniform thickness t . This block is divided into a large number of very small blocks. Each of these blocks will be having a weight w_i . These weights which are acting downward form a system of parallel forces. Consider one such element from the block whose weight is w_i . This force will be acting from the centre of the element. **The co-ordinates of the centre of this element be (x_i, y_i) .**



The centre of the system of parallel gravity forces is known as centre of gravity and can be found out by

$$x_c = \frac{\Sigma F_i x_i}{\Sigma F_i} = \frac{\Sigma w_i x_i}{\Sigma w_i} \gg \gg y_c = \frac{\Sigma F_i y_i}{\Sigma F_i} = \frac{\Sigma w_i y_i}{\Sigma w_i}$$

If ΔA_i is the cross-sectional area of the element and ρ is the density of the material, then

$$w_i = \Delta A_i t \rho g$$

Therefore,

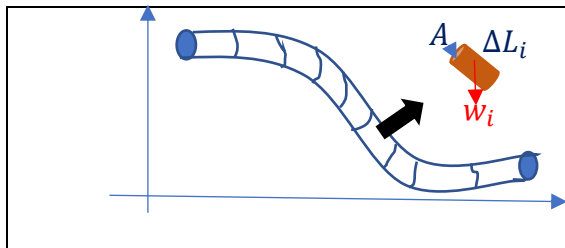
$$x_c = \frac{\Sigma w_i x_i}{\Sigma w_i} = \frac{\Sigma (\Delta A_i t \rho g) x_i}{\Sigma (\Delta A_i t \rho g)} = \frac{t \rho g \Sigma \Delta A_i x_i}{t \rho g \Sigma \Delta A_i} = \frac{\Sigma \Delta A_i x_i}{\Sigma \Delta A_i}$$

Similarly,

$$y_c = \frac{\Sigma \Delta A_i y_i}{\Sigma \Delta A_i}$$

The centre of gravity of shapes with small uniform thickness can be seen now depending on the shape of the middle section of the block. In such case, it is called the centroid of the section (the term centre of gravity is used for three dimensional bodies and centroid for planar shapes). If we divide the block in to infinitesimally small elements of cross-sectional area dA , then the summation can be replaced by integration.

$$x_c = \frac{\int x_i dA}{\int dA} \quad \langle\langle\rangle\rangle \quad y_c = \frac{\int y_i dA}{\int dA}$$



Consider a wire of length L of uniform cross-sectional area A . It is divided into a number of small elements of weights w_i and lengths ΔL_i . These weights which are acting downward will form a system of parallel forces. If one such element is considered, whose centre is at (x_i, y_i) .

$$w_i = \Delta L_i A \rho g$$

The centre of these parallel gravity forces is given by

$$x_c = \frac{\sum w_i x_i}{\sum w_i} = \frac{\sum (\Delta L_i t \rho g) x_i}{\sum (\Delta L_i t \rho g)} = \frac{t \rho g \sum \Delta L_i x_i}{t \rho g \sum \Delta L_i} = \frac{\sum \Delta L_i x_i}{\sum \Delta L_i}$$

Similarly,

$$y_c = \frac{\sum \Delta L_i y_i}{\sum \Delta L_i}$$

The centre of gravity of slender wires with uniform cross section depends on the shape of the axis of the wire.

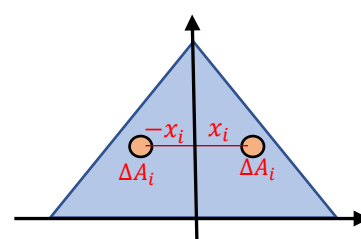
If we divide the slender wire in to infinitesimally small elements of lengths dL , then the summation can be replaced by integration.

$$x_c = \frac{\int x_i dL}{\int dL} \quad \langle\langle\rangle\rangle \quad y_c = \frac{\int y_i dL}{\int dL}$$

If a plane figure has an axis of symmetry, the centroid will lie on that axis.

Consider a shape symmetrical about the y -axis as shown. For every elemental area ΔA_i on one side of this symmetrical axis whose centre is at $+x_i$, there will be a corresponding element on the other side whose centre is at $-x_i$. When we find $\sum \Delta A_i x_i$, it will be equal to zero. Therefore, $x_c = 0$. That is, the centroid lies on the y -axis.

Similarly, if a shape has two axes of symmetry, the centroid will fall on the point of intersection of the two axes.



Reference

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2. Mechanics for Engineers: Beer & Johnston, McGraw Hill
3. Engineering Mechanics: Merriam & Kraig, Wiley
4. Engineering Mechanics: Hibbeler, Pearson

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